Quantum Gravity Dynamics

Chapter 4: Natural-Invariant Dynamics
and the Physical Meaning of Gravity

QGD–098 research edition — referee-verified radial Dirac audit and semiclassical benchmark

QGD research manuscript — independent of the archived historical Chapter 4

26 October 2026 — QGD–101 integration, QGD–105 audit correction, QGD–110 quantum structure, QGD–111/112/113 spin-$\frac{1}{2}$ updates

Abstract

What does the natural-invariant description of gravitation add to the Einsteinian metric equations? We formulate a precise first answer, without postulating a new wave law. Given the composite QGD metric of Chapters 1–2 and the effective gravitational and matter action used in Chapters 2–3, variation with respect to the independent natural-invariant one-forms produces an exact projected Euler equation. The historical three-channel idea thereby admits an unambiguous *action-defined successor*, $\mathscr{S} = \mathscr{G} + \mathscr{T} + \mathscr{Q} = 0$. In the derivative-free, metric-only sector, $\mathscr{Q} = 0$ and the remaining terms are projections of the Einstein–matter residual. On a regular full-rank signed-field representation, those projected equations are equivalent to the effective Einstein equations. The result clarifies the physical role of σ_μ, the origin of its representation freedom, and the exact location where genuinely distinct QGD dynamics would have to arise. We incorporate the Chapter 3 breakthroughs: Newtonian calibration of the exterior natural-invariant amplitude; the exact static Killing-energy expansion; a corrected conserved-current bridge; the leading Dirac-derived quantum density-gradient stress; and a no-go theorem for universal power-law exterior corrections produced solely by restructuring a compact spherical source at fixed mass. It does not assert a microscopic wave equation, a derived σ kinetic term, or a new gravitational observable. This short chapter is intended to grow as these questions are solved; it does not alter the historical Chapter 4. The chapter proves a metric-only quantum-equivalence theorem, imports a verified semiclassical massless-Dirac far-field correction, and audits the historical 9/2 force. A QGD–110 section derives the quantum structure of the master metric from the founding principles: a universal quantum sector fixed by the source's quantum stress, a per-wave quantum conformal factor, and the test of the historical derivative term, and the exact flow law of spin-$\frac{1}{2}$ matter in a curved universal metric; it also records Lorentz-covariance and gauge tests of the main equations. The revision history is in Section 14.

Reading key. **D**: derived from stated assumptions; **A**: adopted effective dynamics; **C**: fixed definition or normalization; **P**: proposed physical interpretation not yet an independent empirical result. These labels distinguish logical status, not scientific importance.

1 From the quantum description to effective geometry

General relativity establishes how geometry behaves when coupled to energy and momentum. It does not, by itself, establish whether geometry is fundamental or an effective macroscopic description. QGD investigates a concrete alternative interpretation: the gravitational field is associated with macroscopic quantum structure of matter, represented by gravitational natural-invariant covectors $\sigma_\mu^{(s)}$; their composites define an effective metric (P). The word *natural invariant* is QGD's historical name for these covectors; a covector is not itself a coordinate-invariant scalar.

The essential distinction is between an *ontological* proposal and a *dynamical* result. The first three chapters supply (i) the four-component macroscopic Dirac structure and controlled local spinor ratios, (ii) the master metric and its signed-sector representations, and (iii) Einsteinian effective dynamics and Newtonian calibration in specified limits. None of these statements alone proves that the individual natural-invariant components constitute independent observable particles or waves.

The general QGD metric is retained unchanged:

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_{s=1}^{N} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \quad \varepsilon_s = \pm 1.} \tag{1}$$

Here $T^a{}_\mu$ is the reference coframe, and the derivative operator $\mathcal{Q}_{ab}$ and coefficient λ remain unmatched proposals in the unrestricted theory. We will derive definite equations in the explicitly stated fixed-coframe, $\lambda = 0$, metric-only effective sector and then explain what changes when the full metric map is used.

2 The new Chapter 4 field equation

In this section set $c = 1$. Restore the physical Einstein coupling $8\pi G_N/c^4$ when using SI units. Consider the composite effective action (A)

$$\Gamma[\sigma, \Psi] = S_{\rm EH}[g(\sigma)] + S_{\rm matter}[g(\sigma), \Psi], \qquad S_{\rm EH} = \frac{1}{16\pi G_N} \int \mathrm{d}^4x \sqrt{-g}\,(R - 2\Lambda). \tag{2}$$

Boundary variations are compactly supported or supplied with the appropriate gravitational boundary term. The matter equations are varied independently. Define $T^{\mu\nu}$ by $\delta S_{\rm matter} = \frac{1}{2}\int \sqrt{-g}\, T^{\mu\nu}\delta g_{\mu\nu}$ for on-shell matter variations, and let

$$\boxed{\mathcal{E}_g^{\mu\nu} := \frac{1}{\sqrt{-g}} \frac{\delta \Gamma}{\delta g_{\mu\nu}} = -\frac{1}{16\pi G_N}\left(G^{\mu\nu} + \Lambda g^{\mu\nu} - 8\pi G_N T^{\mu\nu}\right).} \tag{3}$$

This is a definition of the effective metric Euler derivative, not an independent QGD field equation.

Write the metric-only signed specialization locally as

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \sum_s s_s \sigma_\mu^{(s)} \sigma_\nu^{(s)}, \qquad s_s := -\varepsilon_s \in \{+1, -1\}. \tag{4}$$

All variations below treat the covectors as independent components in a fixed frame and hold the background coframe fixed.

Theorem 2.1 (Action-defined natural-invariant equation (D|A))**.** *For the action* (2) *and metric map* (4)*, the complete Euler derivative of each independent natural-invariant covector satisfies*

$$\boxed{\mathscr{S}^\rho_{(s)} := \frac{1}{\sqrt{-g}} \frac{\delta \Gamma}{\delta \sigma_\rho^{(s)}} = 2 s_s \mathcal{E}_g^{\rho\nu} \sigma_\nu^{(s)} = 0.} \tag{5}$$

Equivalently, with action-defined *channels,*

$$\boxed{\mathscr{S}^\rho_{(s)} = \underbrace{-\frac{s_s}{8\pi G_N}\left(G^{\rho\nu} + \Lambda g^{\rho\nu}\right)\sigma_\nu^{(s)}}_{\mathscr{G}^\rho_{(s)}} + \underbrace{s_s T^{\rho\nu}\sigma_\nu^{(s)}}_{\mathscr{T}^\rho_{(s)}} + \underbrace{0}_{\mathscr{D}^\rho_{(s)}} = 0.} \tag{6}$$

Proof. Vary the signed metric while holding other sectors fixed: $\delta g_{\mu\nu} = s_s(\delta\sigma_\mu^{(s)}\sigma_\nu^{(s)} + \sigma_\mu^{(s)}\delta\sigma_\nu^{(s)})$. Symmetry of $\mathcal{E}_g^{\mu\nu}$ gives $\delta\Gamma = \int \sqrt{-g}\, 2s_s\mathcal{E}_g^{\rho\nu}\sigma_\nu^{(s)}\delta\sigma_\rho^{(s)}$. Equation (5) follows from independent arbitrary compactly supported variations. Substituting (3) proves (43). □

Remark 2.2 (What the three symbols mean)**.** *$\mathscr{G}$ is the effective* geometric-action contribution*, $\mathscr{T}$ is the* matter-stress contribution*, and $\mathscr{Q}$ denotes additional terms arising from a* specified *derivative-dependent metric map when such a map has actually been defined and varied. Thus $\mathscr{Q} = 0$ in the metric-only algebraic sector. The letter Q does* not *by itself mean "quantum". In particular, a genuine quantum matter-loop correction to a metric-only effective action modifies $\mathcal{E}_g$, hence the geometric/matter balance, without necessarily producing any separate $\mathscr{Q}$ channel.*

3 What changes in the unrestricted derivative-dependent metric map

The master proposal (1) permits a nonzero term $-\lambda\mathcal{Q}_{ab}$. Its functional form and coefficient are currently unmatched; the existence of a symbol $\mathcal{Q}_{ab}$ does not establish the independent field equation $\Box_g\sigma_\mu$ or any particular force correction. A conditional chain rule can, however, be proved *before* choosing that operator.

Proposition 3.1 (Functional chain rule for a first-derivative metric map)**.** *Hold the reference coframe fixed and suppose the composite metric is a smooth local function $g_{\mu\nu}(\sigma, \partial\sigma)$ of the independent natural-invariant fields and their first coordinate derivatives. Assume the matter and gravitational action depend on σ only through the resulting metric, and its metric Euler variation is $\delta\Gamma = \int \mathrm{d}^4x\,\sqrt{-g}\,\mathcal{E}_g^{\mu\nu}\,\delta g_{\mu\nu}$ after the appropriate boundary treatment. Then, for arbitrary compactly supported $\delta\sigma$, the natural-invariant Euler derivative obeys*

$$\boxed{\frac{\delta\Gamma}{\delta\sigma_\rho^{(s)}} = \sqrt{-g}\,\mathcal{E}_g^{\mu\nu}\frac{\partial g_{\mu\nu}}{\partial\sigma_\rho^{(s)}} - \partial_\alpha\left(\sqrt{-g}\,\mathcal{E}_g^{\mu\nu}\frac{\partial g_{\mu\nu}}{\partial(\partial_\alpha\sigma_\rho^{(s)})}\right).} \tag{7}$$

If g depends on second or higher derivatives of σ, the corresponding alternating higher total derivatives are required. Direct nonmetric σ couplings add their own *Euler functional derivatives.*

Proof. Use the ordinary chain rule on the specified local metric map, substitute into the metric Euler variation, and integrate by parts the term proportional to $\partial_\alpha\delta\sigma_\rho^{(s)}$. Compact support removes the boundary contribution. A second integration-by-parts argument handles second derivatives, and so on. □

This equation expresses *how* a well-defined derivative metric operator would enter; it does not determine an independent kinetic coefficient from the already accepted action. Without the actual $\mathcal{Q}_{ab}$ and its admissible boundary terms, no numerical quantum-force coefficient can be assigned to this contribution.

Remark 3.2 (The derivative channel on Chapter 2's constructed branch (QGD–101/105))**.** *For both trial completions of $\mathcal{Q}$ in Chapter 2 and any $\lambda \neq 0$, Chapter 2 constructs a particular nine-sector branch on which the metric map has a differential right inverse (Theorems on the nine-sector lift). On that branch, a metric-mediated action gives exactly the effective metric equations, whatever λ is. As Chapter 2 states, this concerns a particular spanning branch, not all field configurations; on other branches, or with σ dependence not passing through g, the question remains open.*

4 Does this equation say anything beyond GR?

One projected field equation per covector need not contain all ten independent symmetric metric equations. What matters is the rank of the map from independently variable signed covectors to the ten metric components.

Proposition 4.1 (Exact regular signed section (D))**.** *In a fixed local reference frame let* $h_{\mu\nu} = g_{\mu\nu} - \bar{g}_{\mu\nu}$ *be any real symmetric* 4×4 *matrix. For any constant* $a > 0$ *set*

$$\boxed{P = aI_4 + \frac{h}{4a}, \qquad N = aI_4 - \frac{h}{4a}.} \tag{8}$$

The four rows of P *and four rows of* N *define eight signed covectors, and*

$$\boxed{P^{\mathsf{T}}P - N^{\mathsf{T}}N = h.} \tag{9}$$

For every symmetric variation δh*, the choice* $\delta P = \delta h/(4a)$ *and* $\delta N = -\delta h/(4a)$ *yields* $\delta g = \delta h$*. Thus the metric differential has rank ten on this section.*

Proof. Expand both products in (9); the $a^2 I$ and $h^2/(16a^2)$ terms cancel, leaving h. Differentiation gives the displayed right inverse. □

Theorem 4.2 (Conditional equivalence with Einsteinian dynamics (D|A))**.** *Suppose the full effective action depends on the signed covectors* only through $g(\sigma)$*, and the admissible covector variations possess a local right inverse to arbitrary symmetric metric variations, as on the section* (8)*. Then*

$$\boxed{\mathscr{S}^{\rho}_{(s)} = 0 \text{ for every sector } s \quad \Longleftrightarrow \quad \mathcal{E}_g^{\mu\nu} = 0.} \tag{10}$$

Thus, on this regular full-rank branch, the action-derived natural-invariant equations are equivalent to the specified *effective Einstein–matter equations.*

Proof. If $\mathcal{E}_g = 0$, every projected Euler derivative is zero. Conversely, let every covector Euler derivative vanish. Use the differential right inverse of Proposition 4.1 to express any compactly supported symmetric metric test variation as an admissible signed-covector variation. Then $\delta\Gamma = 0$ for all metric tests and the fundamental lemma of the calculus of variations gives $\mathcal{E}_g = 0$. □

Interpretation. Writing the Einsteinian equations using natural invariants is a nontrivial local representation and an alternative choice of variables. It does *not*, by itself, generate additional observables beyond GR. In the metric-only sector the transformation $S \mapsto O(x)S$ with $S = \binom{P}{N}$ and $O^{\mathsf{T}}\mathrm{diag}(I_4, -I_4)O = \mathrm{diag}(I_4, -I_4)$ leaves g unchanged. The resulting local $O(4,4)$ freedom is a parametrization redundancy wherever it is a symmetry of the complete action and matter couplings. A physical σ representative need not be unique; adding direct interactions could change this symmetry.

5 The calibrated gravitational field and the normalization distinction

Chapter 3 independently varies both components $A(r)$ and $B(r)$ of the declared classical spherical one-form $\sigma = A\,\mathrm{d}x^0 + B\,\mathrm{d}r$, without setting $A = B$ in the action. On its nondegenerate, asymptotically flat vacuum branch, the adopted Einstein–Hilbert sector gives

$$\boxed{A(r)^2 = B(r)^2 = \frac{C_g}{r}, \qquad C_g = \frac{2G_N M}{c^2} =: r_s.} \tag{11}$$

The value of C_g is fixed *once* by the Newtonian exterior acceleration. It is not derived from the microscopic Dirac equation; its calibration is legitimate correspondence with the observed Newtonian mass parameter, as in the historical Einstein normalization. No other coefficient is adjusted to recover higher orders.

When an independently selected reference fermion mass $m_{\rm ref} > 0$ supplies a reduced Compton length $\bar{\lambda}_{\rm ref} = \hbar/(m_{\rm ref}c)$, Chapter 3 defines the dimensionless *geometric inverse-strength convention*

$$\boxed{|\alpha_G^{\rm geom}|^2 := \frac{\bar{\lambda}_{\rm ref}}{C_g} = \frac{\hbar c}{2G_N M m_{\rm ref}}.} \tag{12}$$

The original common spinor multiplier $\alpha_G^{\rm spin}$ in $\Psi = (\alpha_G^{\rm spin}/r)\sum_a \psi_a u_a e^{\pm iS_a/\hbar}$ is *not* thereby normalized from the geometric equations. Indeed, the simultaneous change $\alpha_G^{\rm spin} \mapsto z\alpha_G^{\rm spin}$, $\psi_a \mapsto \psi_a/z$ with $z \neq 0$ leaves the physical state unchanged. Its absolute normalization comes from an independently specified current or occupied-state convention. In particular, $\alpha_G^{\rm spin} = \alpha_G^{\rm geom}$ is not a theorem of Chapters 1–3.

The observable static Killing-energy potential per unit held test mass is

$$\boxed{\Phi_{\rm stat}(r) = c^2\left(\sqrt{1 - \frac{C_g}{r}} - 1\right), \qquad r > C_g.} \tag{13}$$

Its convergent, wholly *classical* expansion is

$$\boxed{\Phi_{\rm stat}(r) = -\frac{G_N M}{r} - \frac{G_N^2 M^2}{2c^2 r^2} - \frac{G_N^3 M^3}{2c^4 r^3} - \frac{5G_N^4 M^4}{8c^6 r^4} + \cdots.} \tag{14}$$

All powers of $\hbar$ cancel when (12) is used merely as a parametrization of C_g. Thus these higher orders are not evidence of quantum-corrected vacuum gravity.

6 The microscopic Dirac bridge: what it fixes

For one locally on-shell, positive-energy, outward-propagating Dirac WKB branch with conserved particle flux $\dot{N} > 0$, Chapter 1 and Chapter 3 give the magnitude-only current identity

$$\boxed{|\alpha_G^{\rm spin}\psi_0(r)|^2 = \frac{\dot{N}\,\Delta(r)}{8\pi c^2 p(r)}, \qquad \Delta = E_{\rm loc} + m_{\rm eff}c^2, \quad p > 0.} \tag{15}$$

This is a transport constraint. It determines the envelope only relative to an independently fixed momentum profile and flux; it does *not* determine a $1/r$ gravitational potential, the geometric normalization, or an absolute phase. A static bound source with zero net radial current is *not* normalized using this outgoing flux formula. Its spinor profiles must be obtained from its own coupled boundary-value equations with conserved occupation.

7 A controlled quantum matter contribution to the source

To make the $\hbar$ dependence precise without silently inserting a new gravitational operator, use a stated positive-energy, fixed-spin, nonrelativistic Dirac-to-Pauli limit and write $\varphi = \sqrt{n}\,e^{iS/\hbar}\chi$ with constant unit Pauli spinor χ, $n > 0$, and fermion mass $m_f > 0$. The polar identity for the Schrödinger kinetic energy is

$$\boxed{E_{\rm kin}[n,S] = \int {\rm d}^3x \left[\frac{n|\boldsymbol{\nabla}S|^2}{2m_f} + \frac{\hbar^2}{8m_f}\frac{|\boldsymbol{\nabla}n|^2}{n}\right].} \tag{16}$$

The quantum density-gradient functional and associated momentum-flux tensor (one admissible improvement convention) satisfy

$$\boxed{Q[n] = -\frac{\hbar^2}{2m_f}\frac{\nabla^2\sqrt{n}}{\sqrt{n}}, \qquad \Pi^Q_{ij} = \frac{\hbar^2}{4m_f}\left(\frac{\partial_i n \partial_j n}{n} - \partial_i\partial_j n\right), \qquad \partial_j \Pi^Q_{ij} = n\partial_i Q.} \tag{17}$$

This is a derived coefficient of *quantum matter stress*, not a derived coefficient of a standalone $\hbar^2(\nabla\sigma)^2$ gravitational kinetic term. To leading nonrelativistic order, the *adopted* sourced Einstein–Hilbert equations reduce to Poisson's law:

$$\boxed{\nabla^2\Phi = 4\pi G_N m_f n, \qquad \mathcal{U} := -\frac{2\Phi}{c^2}, \qquad \nabla^2\mathcal{U} = -\frac{8\pi G_N m_f}{c^2} n.} \tag{18}$$

If the physical source profile $n(x;\hbar)$ is independently determined from its quantum dynamics, its $\hbar$ dependence influences the sourced metric and consequently the corresponding temporal natural-invariant *combination*. This is a source effect, not a new vacuum propagation law. Relativistic stress, pressure, spin, multi-orbital occupations and binding require the complete Dirac–Einstein source instead of simply substituting one n.

Proposition 7.1 (Compact spherical exterior at fixed mass (D|A))**.** *Let $\rho_\hbar(r)$ be an integrable, nonnegative spherically symmetric mass density with fixed total mass $M = 4\pi\int_0^\infty \rho_\hbar(r) r^2\,\mathrm{d}r$, compactly supported in $r \le R$. Assume Poisson's equation and $\Phi(\infty) = 0$. Then $\Phi(r) = -G_N M/r$ and $F_r = -G_N M m_t/r^2$ for every $r > R$, independently of the interior $\hbar$ dependence.*

Proof. Spherical Poisson integration gives $r^2\Phi'(r) = 4\pi G_N \int_0^r \rho_\hbar(s) s^2\,\mathrm{d}s$. The integral equals $M/(4\pi)$ outside the support. Integrate once more with the stated asymptotic boundary condition. □

Thus a universal exterior term proportional to $G_N M\hbar^2/(m_f^2 c^2 r^4)$ does *not* follow from changing only the profile of a compact spherical quantum source while keeping its mass and the Poisson law fixed. Noncompact sources, altered total mass, non-spherical multipoles or independently derived quantum field equations are distinct questions.

Remark 7.2 (Varying spin (QGD–113))**.** *The fixed-spin assumption above is a special case. When the spin direction $\hat{s}$ varies, the exact spin-$\frac{1}{2}$ flow law of Section 11.5 adds the spin quantum potential $(\hbar^2/8m_f)|\nabla\hat{s}|^2$ and a spin-vorticity force; both vanish for constant $\hat{s}$, recovering Eq. (17).*

8 Status of the historical quantum-force candidate

The proposed universal attractive force

$$\boxed{F_{r,\mathrm{hist}} = -\frac{G_N M m_t}{r^2}\left[1 + \frac{9}{2}\left(\frac{\hbar}{m_f c r}\right)^2 + \cdots\right]} \tag{19}$$

remains a historical *candidate*, not a conclusion of the accepted action. The reciprocal of $e^{2im_f cr/\hbar}$ is $e^{-2im_f cr/\hbar}$: inverting nonconstant Taylor terms separately does not invert the complete exponential. The historical cubic energy also gives a sign opposite to an enhancement of attraction.

A first-order weak-field expansion of a specified Liouville-normal, separated Schwarzschild radial Dirac equation produces coefficients depending on its angular channel κ. Their derivatives are not coordinate-, tetrad-, or field-redefinition-invariant force observables. In particular the lowest $\kappa = -1$ channel does not produce the historical term at order $G_N\hbar^2/r^4$ in that normal form. Nothing here establishes a universal quantum change to the exterior gravitational acceleration. To claim one, first derive a covariant $\hbar$-dependent action, vary it including all connection and metric dependence, and calculate a gauge-/representation-invariant physical observable.

8.1 Independent referee test of the radial Dirac coefficient (QGD–098)

The following is a *conditional theorem about a quantum test field*. It does not modify the natural-invariant field equations. Adopt the classical, Newton-calibrated Schwarzschild exterior of Eq. (11), minimally couple a massive four-component Dirac field through the induced orthonormal-frame spin connection, and use the standard separated, fixed-tetrad radial basis. This additional coupling and the reduction to a single radial mode are assumptions of the *test-field model*, not consequences of the flat-space spinor-component ratios alone. Keep $r > r_s > 0$ and define

$$\ell_C := \frac{\hbar}{m_f c}, \quad x := \frac{r}{\ell_C}, \quad a := \frac{r_s}{\ell_C}, \quad e := \frac{E}{m_f c^2}, \quad F := 1 - \frac{a}{x}, \quad \frac{\mathrm{d}}{\mathrm{d}x_*} := F\frac{\mathrm{d}}{\mathrm{d}x}, \tag{20}$$

where $\kappa \in \mathbb{Z} \setminus \{0\}$ is the spinor angular index. For the stated radial spinor gauge the first-order field equations read

$$\boxed{\begin{aligned} \frac{\mathrm{d}f}{\mathrm{d}x_*} &= (e + \sqrt{F})g - \frac{\kappa\sqrt{F}}{x}f, \\ \frac{\mathrm{d}g}{\mathrm{d}x_*} &= \frac{\kappa\sqrt{F}}{x}g + (\sqrt{F} - e)f. \end{aligned}} \tag{21}$$

Proposition 8.1 (Liouville-normal radial coefficient (D|A))**.** *Assume Eqs. (20)–(21), $e = 1$, and $a/x < 1$. Put $B := 1 + \sqrt{F}$, $K := \kappa\sqrt{F}/x$, and*

$$P := \frac{F_x}{F} - \frac{B_x}{B}, \qquad Q := FK_x - F\frac{B_x}{B}K - K^2 + 1 - F. \tag{22}$$

Then elimination of g gives

$$f_{xx} + Pf_x + \frac{Q}{F^2}f = 0. \tag{23}$$

After the invertible local Liouville rescaling $f = \exp(-\frac{1}{2}\int P\,\mathrm{d}x)u$, the transformed equation is

$$\boxed{u_{xx} + \mathcal{V}_\kappa(x, a)u = 0, \qquad \mathcal{V}_\kappa = \frac{Q}{F^2} - \frac{1}{2}P_x - \frac{1}{4}P^2.} \tag{24}$$

Its expansion at fixed $x > 0$ about $a = 0$ is

$$\boxed{\begin{aligned} \mathcal{V}_\kappa = &-\frac{\kappa(\kappa+1)}{x^2} + \frac{a}{x} + \frac{3 - 4\kappa^2 - \kappa}{4}\frac{a}{x^3} \\ &+ \frac{2a^2}{x^2} + \frac{-64\kappa^2 + 4\kappa + 69}{64}\frac{a^2}{x^4} + O(a^3). \end{aligned}} \tag{25}$$

In particular,

$$\boxed{\mathcal{V}_{-1}(x, a) = \frac{a}{x} + \frac{2a^2}{x^2} + \frac{a^2}{64x^4} + O(a^3),} \tag{26}$$

so the $O(a/x^3)$ correction vanishes in the $\kappa = -1$ channel. The displayed expansions are local analytic expansions under the stated hypotheses; they are not uniform near $r = r_s$.

Proof. Solve the first equation in (21) for $g = (Ff_x + Kf)/B$. Differentiate with respect to x_*, substitute the second equation, and collect the f_x and f coefficients to obtain Eq. (23). Direct differentiation of the Liouville rescaling gives Eq. (24). Using $\sqrt{1 - a/x} = 1 - a/(2x) - a^2/(8x^2) + O(a^3)$ in the resulting *exact* coefficient gives Eq. (25). Setting $\kappa = -1$ proves Eq. (26). □

Remark 8.2 (What is and is not a quantum force)**.** *At $e = 1$ one may* formally *rewrite Eq. (24) as a zero-energy one-dimensional Schrödinger equation, with $U_{\mathrm{radial}} := -m_f c^2 \mathcal{V}_\kappa/2$. Its $O(a/x^3)$ contribution is*

$$\delta U_\kappa = -\gamma_\kappa \frac{G_N M \hbar^2}{m_f c^2 r^3}, \qquad \gamma_\kappa := \frac{3 - 4\kappa^2 - \kappa}{4}. \tag{27}$$

Differentiating this representation-dependent radial term *yields* $-3\gamma_\kappa G_N M\hbar^2/(m_f c^2 r^4)$*, but this is* ***not*** *by itself a local force observable or an invariant correction to the gravitational metric. In particular,* $\gamma_{-1} = 0$ *and* $\gamma_{+1} = -1/2$*. The historical* attractive *coefficient* $9/2$ *would require* $\gamma_\kappa = 3/2$*, i.e.* $4\kappa^2 + \kappa + 3 = 0$*, which has no real root. The* $a^2/(64x^4)$ *term at* $\kappa = -1$ *is likewise an* effective radial-equation coefficient*, not a new vacuum Einstein tensor. It must not be reported as a distinct QGD quantum-gravitational force.*

Referee/proof boundary. The result above rests on a stated classical Schwarzschild exterior, the minimally coupled test Dirac equation, a fixed tetrad and spinor angular basis, and a specified threshold energy $e = 1$. It therefore follows from Chapters 1–2 only *after* their declared effective metric/Dirac coupling is adopted. It is not derived solely from the flat-space macroscopic Dirac ratios. Exact symbolic differentiation verifies Eqs. (23)–(26); this is not a Lean-kernel proof. A physical quantum-gravity claim would additionally require the modified, conserved metric/source field equations and an invariant observable, with no further fit to the historical $9/2$ coefficient.

9 A published semiclassical Dirac benchmark in natural-invariant variables

The historical $9/2$ Compton-scale force is not established by the original phase manipulation. There is, however, a distinct and quantitatively fixed $\hbar$-dependent *gravitational* correction that can be imported into the effective metric sector. Carlson, Anderson, Fabbri, Fagnocchi, Hirsch and Klyap computed the leading far-field renormalized stress tensor of one free *massless four-component* Dirac field in an asymptotically empty (Boulware) state, and checked its gravitational backreaction against previous one-loop calculations [7]. This is a benchmark of semiclassical Einstein–Dirac physics, not an independent derivation of gravitational dynamics from the original flat-space QGD Dirac action.

Proposition 9.1 (Massless Dirac far-field gravitational correction (D|A))**.** *Adopt the metric-only effective Einstein–Hilbert action with a minimally coupled quantized free massless four-component Dirac field, the renormalized Boulware state, asymptotic flatness, fixed ADM mass* $M > 0$*, areal radius* r*, and leading-order semiclassical backreaction. In units* $G_N = c = \hbar = 1$*, the published leading stress tensor is*

$$\boxed{\langle T^{\mu}{}_{\nu}\rangle_{\rm ren} = \frac{M}{40\pi^2 r^5}\,\mathrm{diag}(4,2,-3,-3) + O(r^{-6}).} \tag{28}$$

For the far-field static metric

$$\mathrm{d}s^2 = -f(r)\mathrm{d}t^2 + \left(1 - \frac{2\mathfrak{m}(r)}{r}\right)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2, \qquad f(r) = \left(1 - \frac{2\mathfrak{m}(r)}{r}\right)e^{2\nu(r)}, \tag{29}$$

the asymptotically normalized functions satisfy

$$\mathfrak{m}(r) = M\left(1 + \frac{1}{5\pi r^2}\right) + O(r^{-3}), \tag{30}$$

$$\nu(r) = \frac{M}{15\pi r^3} + O(r^{-4}), \tag{31}$$

$$f(r) = 1 - \frac{2M}{r} - \frac{4M}{15\pi r^3} + O(r^{-4}), \tag{32}$$

where in the last expression $O(r^{-4})$ *also includes mixed classical–semiclassical orders. Restoring*

$G_N, c, \hbar$, *the* leading far-field *supported-static-body Killing-energy potential per unit test mass is*

$$\boxed{\begin{aligned}\Phi^{\mathrm{sc}}_{\mathrm{QGD}}(r) = c^2\left(\sqrt{1-\frac{2G_N M}{c^2 r}}-1\right)\\ -\frac{2G_N^2 M\hbar}{15\pi c^3 r^3}+O\!\left(\frac{G_N^3 M^2\hbar}{c^5 r^4}\right)+O(\hbar^2),\end{aligned}} \tag{33}$$

where the final $O(\hbar^2)$ denotes omitted higher quantum orders, not a uniform SI-dimensional bound. The displayed quantum coefficient is meaningful in the far field $r \gg r_s = 2G_N M/c^2$, where the quantum backreaction is perturbatively small. It must not be extrapolated unchanged to $r \simeq r_s$.

Proof. For the ansatz (29), the radial Einstein equations at leading quantum order read

$$\mathfrak{m}' = -4\pi r^2 \langle T^t{}_t\rangle_{\mathrm{ren}}, \qquad \nu' = \frac{4\pi r(\langle T^r{}_r\rangle_{\mathrm{ren}} - \langle T^t{}_t\rangle_{\mathrm{ren}})}{1-2\mathfrak{m}/r}. \tag{34}$$

Substituting (28) gives $\mathfrak{m}' = -2M/(5\pi r^3)$ and $\nu' = -M/(5\pi r^4) + O(r^{-5})$. Integrating with $\mathfrak{m}(\infty) = M$ and $\nu(\infty) = 0$ proves (30)–(31). Consequently $f = 1-2M/r-4M/(15\pi r^3)+O(r^{-4})$. For an observer held static in this exterior, $\Phi = c^2(\sqrt{f}-1)$; expanding to first order in the renormalized Dirac stress and restoring dimensions proves (33). This is a sourced asymptotic expansion, not the solution of a freely chosen scalar force law. □

Remark 9.2 (Quantum gravitational force and Planck-scale residue)**.** *The* quantum part *of the signed radial force on a held static test mass m_t has leading far-field behavior*

$$\boxed{\delta F_r^{\mathrm{sc}} = -\frac{2G_N^2 M m_t \hbar}{5\pi c^3 r^4} + \textit{higher orders.}} \tag{35}$$

Relative to the Newtonian attraction, its leading magnitude is $\frac{2}{5\pi}\frac{G_N\hbar}{c^3 r^2}$, which is governed by the Planck length squared, not the inverse square of an arbitrarily selected probe mass. It is therefore not *the historical $(9/2)(\hbar/(m_f c r))^2$ correction. The classical piece $-m_t\Phi'_{\mathrm{stat}}$ remains defined by the exact background square root. The quantum term is a controlled far-field correction, not an exact all-radius force.*

Remark 9.3 (Natural-invariant reconstruction and status)**.** *The full signed/tetrad QGD master metric (1) is unchanged. For the nondegenerate one-sector static spherical* metric representation *discussed in Chapter 2, write $\sigma = A(r)\mathrm{d}t + B(r)\mathrm{d}r$ and put $F = 1-A^2$, $D = 1+B^2-A^2$ so that the diagonal radial metric coefficient is D/F. Reconstruct the semiclassical metric by $A^2 = 1-f$ and $D/F = (1-2\mathfrak{m}/r)^{-1}$. Then, in the far field and to the first displayed quantum order,*

$$\boxed{\begin{aligned}A^2(r) &= \frac{2M}{r}+\frac{4M}{15\pi r^3}+O(r^{-4}),\\ B^2(r) &= \frac{2M}{r}+\frac{2M}{5\pi r^3}+O(r^{-4})\end{aligned}} \qquad (G_N = c = \hbar = 1). \tag{36}$$

Thus the familiar equality $A^2 = B^2$ is a property of the classical Schwarzschild branch, not *of the quantum-corrected source. This construction shows that QGD variables can represent the published semiclassical metric; it does not establish a new dynamical law for σ_μ.*

Assumptions and formal-verification boundary. The algebraic implications from (28) and (34) to (33) are suitable for a Lean-style theorem once analytic asymptotic bounds and positivity conditions on the metric are specified. The renormalized-stress coefficient in (28) is a *published quantum field-theoretic input*, not a theorem derived from Chapter 1. The complete original four-mode wavefunction has not been proved to generate this Boulware vacuum as its unique quantum state. At the current metric-only effective level, the equivalence theorem in the next section applies: this correction is already predicted by semiclassical GR with the same Dirac matter.

10 Can QGD predict a quantum force independently of semiclassical GR?

This is the central discriminating question of QGD–096. Here *semiclassical Einstein gravity* means a classical metric sourced by a renormalized expectation value or effective action of quantum matter; *quantized gravitational effective field theory* includes fluctuations of the gravitational field itself. The two comparisons are different. A Dirac wave in a prescribed metric, the quantum-potential term of a Pauli reduction, and an $\hbar$-dependent source profile are not, by themselves, quantum corrections to the gravitational vacuum response. The relevant test is an invariant observable calculated from an effective dynamics which QGD derives but which the selected GR-plus- quantum-matter comparator does not already possess.

Theorem 10.1 (Metric-only quantum-equivalence obstruction (D|A))**.** *Let $\mathcal{F}$ be the algebraic signed-sector map $\sigma \mapsto g[\sigma]$ on a regular branch whose differential is surjective onto symmetric metric variations, with identical boundary conditions and matter states on both sides. Let $\Gamma_{\rm sc}[g,\Psi]$ be a specified semiclassical Einstein–matter effective functional, possibly including renormalized quantum matter loops, and assume*

$$\boxed{\Gamma_{\rm QGD}[\sigma,\Psi] = \Gamma_{\rm sc}[\mathcal{F}(\sigma),\Psi]} \tag{37}$$

with no independent σ interaction or separately quantized σ measure. Then stationary QGD fields on this branch and stationary fields of $\Gamma_{\rm sc}$ determine the same set of physical metrics and metric/matter observables. In particular, no additional universal $\hbar$-dependent gravitational-force coefficient is implied solely by expressing g through σ.

Proof. For any admissible variation, the functional chain rule gives

$$\delta\Gamma_{\rm QGD} = \int \mathrm{d}^4x\,\sqrt{-g}\,\mathcal{E}^{\mu\nu}_{\rm sc}\,\frac{\partial g_{\mu\nu}}{\partial\sigma^{(s)}_\rho}\delta\sigma^{(s)}_\rho + \delta_\Psi\Gamma_{\rm sc}, \tag{38}$$

where repeated sector indices are summed. A stationary metric of $\Gamma_{\rm sc}$ therefore makes the QGD pullback stationary. Conversely, by surjectivity every compactly supported symmetric δg is induced by an admissible $\delta\sigma$; stationarity of the pullback and the independently imposed matter equations imply $\mathcal{E}^{\mu\nu}_{\rm sc} = 0$. Because all specified observables are functionals of the same physical metric and matter state, they agree. The result is an on-shell, mean-effective-action theorem: it makes no claim about quantizing the redundant variables with a new functional measure, about rank-deficient branches, or about a new action not covered by (37). □

Remark 10.2 (What the theorem does, and does not, rule out)**.** *It rules out* new observable coefficients arising from a metric reparametrization alone*. It does not forbid a distinct QGD quantum mechanism. For example, an independently matched covariant correction $\Delta\Gamma_{\rm QGD}[\sigma,\Psi;\hbar]$ could alter the field equations, whether through additional natural-invariant physics, a quantum-derived metric operator, or a consistently specified quantum measure. But a coefficient copied from a known GR effective action, or chosen to reproduce the historical force, would not constitute an independent QGD prediction. A meaningful comparison must specify the same physical source, asymptotic conditions, renormalization prescription, and gauge-invariant observables.*

10.1 An exact exterior test of the historical $9/2$ law

The attractive force candidate in Eq. (19) can be integrated unambiguously at the displayed order. Put $\ell_C = \hbar/(m_f c)$, where the reference species m_f must be specified independently. In the weak-field nonrelativistic interpretation, $F_r = -m_t \partial_r \Phi$ and $\Phi(\infty) = 0$ imply

$$\boxed{\Phi_{9/2}(r) = -\frac{G_N M}{r} - \frac{3 G_N M \ell_C^2}{2 r^3} + \text{higher-order terms.}} \tag{39}$$

For the isolated displayed r^{-3} term and $r > 0$, $\nabla^2(r^{-3}) = 6 r^{-5}$, so

$$\boxed{\nabla^2 \Phi_{9/2}(r) = -\frac{9 G_N M \ell_C^2}{r^5} \neq 0 \quad (r > R),} \tag{40}$$

where R is the support radius of an assumed compact source and higher-order terms have been omitted consistently. If one tried to retain the *unmodified* Poisson equation, the same radial tail would demand an exterior effective density

$$\boxed{\rho_{\text{eff}}(r) = -\frac{9 M \ell_C^2}{4\pi r^5}, \qquad r > R,} \tag{41}$$

which is not the original empty exterior. This is a diagnostic *effective* density, not a claim about negative physical matter. Therefore the historical universal correction cannot be generated solely by changing a compact spherical source profile under the same vacuum Poisson equation; it requires a justified exterior quantum source/response, a modified field equation, or other physics.

At the fully relativistic level, this weak-field calculation is not by itself a complete covariant metric solution. The independently varied Einstein–Hilbert vacuum branch of Chapter 2 already fixes the exact spherical exterior to Schwarzschild. A new universal $1/r^3$ metric perturbation must consequently be checked against the *complete* modified field equations and their conservation and boundary conditions; a scalar force ansatz cannot perform that check. If m_f denotes the test particle mass, its Compton scale also makes the force state/species-dependent, so the claim is not automatically a universal exterior metric correction.

10.2 A falsifiable QGD-specific matching test

A future new term should be expressed as the difference between a microscopically justified QGD effective functional and the specified semiclassical comparator:

$$\boxed{\Delta\Gamma[\sigma, \Psi; \hbar] := \Gamma_{\text{QGD}}[\sigma, \Psi; \hbar] - \Gamma_{\text{sc}}[g[\sigma], \Psi; \hbar].} \tag{42}$$

At fixed physical matter state and boundary data, a nonzero $\Delta\Gamma$ is *necessary but not sufficient* for a new observable: it may contain a boundary term, a redundancy, or an operator removable by permissible redefinitions. The minimum verification gates are:

1. Derive $\Delta\Gamma$ from the stated microscopic/macroscopic Dirac construction or declare any additional physical axiom.

2. Specify its covariant dependence on the full signed master metric, $\sigma^{(s)}$, and any derivative operators; include its independent normalization, units and boundary terms.

3. Vary the *complete* functional, including connection, coframe, stress-energy and possible nonlocal contributions, and check Bianchi/conservation identities and regular branches.

4. Fix at most the permitted Newtonian calibration coefficient, then predict a second observable without re-fitting it.

5. Compare with *both* semiclassical GR plus quantum matter and, where applicable, low-energy quantized-GR effective-field-theory corrections, at matched sources and renormalization convention.

A quantum Dirac radial equation in the unmodified Schwarzschild metric meets none of the first three gates for a new *gravitational field* operator. The previously derived quantum density-gradient stress meets the microscopic matter-side test, but its gravitational consequences within Eq. (2) are also consequences of the chosen semiclassical Einstein–matter system. Hence the present QGD action does *not yet* close the deeper foundational gap of demonstrating that QGD itself predicts a quantum correction independently of semiclassical general relativity.

11 The quantum structure of the master metric (QGD–110)

This section uses only the founding principles: F1, the microscopic Dirac field; F2, coarse-graining and the macroscopic spinor equation; F3, the master metric with its reference term; F4, the adopted Einstein–Hilbert-plus-matter action; F5, the Newtonian calibration. Imported results are marked L.

11.1 A metric-level quantum term and the field equations

Proposition 11.1 (Exact split identity (D))**.** *Write the physical metric as $g = g_{\rm cl} + q$. If $G[g] = 8\pi G\,T$, then identically*

$$G[g_{\rm cl}] = 8\pi G\,(T + T_Q), \qquad T_Q := -\frac{G[g] - G[g_{\rm cl}]}{8\pi G}. \tag{43}$$

A quantum term in the master metric therefore always produces a quantum-corrected Einstein equation for the classical part. By itself this is a rewriting. It acquires content only when F1–F5 fix q, which they do through the quantum stress of the matter.

11.2 The quantum term in the static natural-invariant variables

Proposition 11.2 (Exact identity (D))**.** *In the static branch ($F = 1 - A^2$, $D = 1 + B^2 - A^2$, areal radius),*

$$A^2 - B^2 = 1 - FH, \qquad \frac{\mathrm{d}}{\mathrm{d}r}\ln(FH) = \frac{8\pi r(\rho + p_r)}{1 - 2\mathfrak{m}/r}, \tag{44}$$

with $H = D/F$. Hence $A = B$ exactly whenever $\rho + p_r = 0$: in vacuum, for Maxwell fields and for a cosmological constant.

For the vacuum of the microscopic Dirac field (Proposition 9.1), Eq. (36) gives

$$A^2 - B^2 = -\frac{2}{15\pi}\,\frac{G_N^2 M\hbar}{c^5 r^3}, \tag{45}$$

because vacuum polarization of the microscopic Dirac field has $\rho + p_r < 0$. Inserting this q into Eq. (43) returns $T_Q = -T^{\rm Dirac}$ in all three independent components (verified symbolically). This is the quantum term F1 and F4 place in the master metric. Its scale is ℓ_P^2, not the historical ℓ_Q, and by Theorem 10.1 it coincides with semiclassical GR.

Proposition 11.3 (Test of the historical derivative trial (D))**.** *For the classical exterior sector $A = B = \sqrt{2M/r}$ on the flat spherical reference, the trial $\mathcal{Q}_{ab} = (\bar\nabla_a\sigma^c)(\bar\nabla_b\sigma_c)$ has only angular components, $\mathcal{Q}_{\theta\theta} = 2M/r$, and in areal gauge gives, to first order in M, $\Delta f = 0$ and $\Delta h = -4\lambda M/R^3$. The derived term requires $\Delta f = -4M\ell_P^2/(15\pi R^3)$ and $\Delta h = +2M\ell_P^2/(5\pi R^3)$. No λ matches both, so the trial is not the quantum term of the foundations.*

11.3 A quantum conformal factor derived from the wave equation

Proposition 11.4 (Quantum conformal factor (D))**.** *For a scalar wave $\psi = Re^{iS/\hbar}$ of mass M obeying $(\Box_g - M^2c^2/\hbar^2)\psi = 0$, the real part of the wave equation is exactly the Hamilton–Jacobi equation of a conformally rescaled metric,*

$$\tilde{g}^{\mu\nu}\partial_\mu S\partial_\nu S + M^2c^2 = 0, \qquad \tilde{g}_{\mu\nu} = \Omega_\psi^2 g_{\mu\nu}, \qquad \Omega_\psi^2 = 1 - \lambda_C^2\frac{\Box_g R}{R}, \quad \lambda_C = \frac{\hbar}{Mc}. \tag{46}$$

Wherever $\Omega_\psi^2 > 0$, the integral curves of $\tilde{g}^{\mu\nu}\partial_\nu S$ are timelike geodesics of $\tilde{g}$.

Proof. The real part is $g^{\mu\nu}\partial S\partial S + M^2c^2 = \hbar^2\Box_g R/R$ (verified symbolically). For the second statement, $u_\mu = \partial_\mu S/(Mc)$ is a gradient with $\tilde{g}(u,u) = -1$, so $u^\nu\tilde{\nabla}_\nu u_\mu = \frac{1}{2}\partial_\mu\tilde{g}(u,u) = 0$. □

This is the quantum potential of Chapter 1 (Eq. wkb-quantumHJ) and of Section 7, in geometric form. Calling the integral curves *trajectories* is an interpretation. The theorem itself is a statement about the wave's phase and amplitude.

11.4 How the quantum term enters the master metric

Theorem 11.5 (Two consistent routes, one inconsistent (D|A))**.** *Under F1–F5:*

1. ***Per wave (consistent).*** *Ω_ψ^2 multiplies the universal master metric for that wave (spin-0 matter; spin-$\frac{1}{2}$ matter in Section 11.5), $\tilde{g}^{(\psi)}_{\mu\nu} = \Omega_\psi^2 T^a{}_\mu T^b{}_\nu[\eta_{ab} + \sum_s s_s\sigma_a^{(s)}\sigma_b^{(s)}]$, and governs that wave's phase dynamics. Universal free fall is recovered as $\lambda_C \to 0$; at finite λ_C the mass dependence is that of ordinary quantum mechanics in a gravitational field.*

2. ***Through the source's stress tensor (consistent, universal).*** *On shell the wave's stress tensor splits exactly as*

$$T_{\mu\nu} = \rho\, u_\mu u_\nu + T^Q_{\mu\nu}, \qquad T^Q_{\mu\nu} = \hbar^2\big[\partial_\mu R\partial_\nu R - \tfrac{1}{2}g_{\mu\nu}\big((\partial R)^2 + R\,\Box_g R\big)\big], \tag{47}$$

   with $\rho = M^2c^2R^2$ and the on-shell $u_\mu = \partial_\mu S/(Mc)$. Under F4, T^Q curves spacetime with coupling $8\pi G$, and through Eq. (43) it appears as a universal quantum sector q of the master metric. For a coherent static source of size L, $T^Q_{00}/\rho c^2 = O(\lambda_C^2/L^2)$.

3. ***Source's conformal factor in the universal metric (inconsistent).*** *For $g = (1+\varepsilon)g_{\rm cl}$, the linearized Einstein tensor $G_{00} = -\nabla^2\varepsilon$ carries no factor of G; for a Gaussian amplitude it equals $6\lambda_C^2/L^4$. This contradicts F4 at order $\hbar^2G^0$, and would require replacing F4.*

The result is a two-level master metric derived from the founding principles:

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu\Big[\eta_{ab} + \sum_s s_s\sigma_a^{(s)}\sigma_b^{(s)} + q_{ab}\Big], \qquad \tilde{g}^{(\psi)}_{\mu\nu} = \Omega_\psi^2\, g_{\mu\nu},} \tag{48}$$

with sector signs $s_s = -\varepsilon_s$ in the notation of Eq. (1), and where q is fixed by F4 from the quantum stress of the source (vacuum polarization, Eq. (45), and the coherent-wave stress T^Q), and Ω_ψ by Eq. (46). This gives the founding master metric's quantum slot, historically $-\lambda\mathcal{Q}$, a derived occupant. As $\hbar \to 0$, q's quantum part and $\Omega_\psi - 1$ vanish, recovering the classical restriction and every classical result of Chapters 2–4. The structure reproduces quantum mechanics in curved spacetime and semiclassical gravity, and it does not quantize the metric: the limitations of semiclassical gravity, for example for macroscopic superpositions of a source, apply unchanged.

Proposition 11.6 (No natural-invariant dynamics beyond g under F1–F5 (D))**.** *The microscopic Dirac action uses a vierbein of g and is locally Lorentz invariant (Chapter 2, Clifford compatibility), so it depends only on g. The Maxwell, scalar-contact and Einstein–Hilbert terms and the quantum matter effective action $\Gamma[g]$ are functionals of g. Theorem 10.1 therefore applies: dynamics acting on more than g requires an additional founding principle.*

11.5 The per-wave law for spin-$\frac{1}{2}$ matter (QGD–112, QGD–113)

For spin-$\frac{1}{2}$ matter no metric conformal to g makes the flow lines geodesic (QGD–111). The exact replacement follows from the macroscopic spinor equation alone. Define $\sigma = \bar{\psi}\psi \neq 0$, the convective momentum $P_\mu = -\operatorname{Im}(\bar{\psi}\nabla_\mu\psi)/\sigma$, the spin remainder $\Lambda = \nabla\bar{\psi}\cdot\nabla\psi - |\bar{\psi}\nabla\psi|^2/\sigma$, the spin tensor $S^{ab} = \bar{\psi}\Sigma^{ab}\psi$ with $\Sigma^{ab} = \frac{i}{2}[\gamma^a, \gamma^b]$, and the spin vorticity $F = \mathrm{d}P$.

Theorem 11.7 (Spin-$\frac{1}{2}$ flow law in a curved universal metric (D))**.** *In Chapter 1's conventions,* $(\gamma^\mu\nabla_\mu)^2 = -\Box + \frac{1}{4}R$ *and* $[\nabla_\mu, \nabla_\nu] = -\frac{1}{4}R_{ab\mu\nu}\gamma^a\gamma^b$ *on spinors. Consequently, exactly,*

$$P\cdot P + M^2 + \tfrac{1}{4}R = \frac{\Box\sqrt{\sigma}}{\sqrt{\sigma}} - \frac{\Lambda}{\sigma}, \qquad F_{\mu\nu} = -\frac{\operatorname{Im}(\nabla_{[\mu}\bar{\psi}\nabla_{\nu]}\psi)}{\sigma} - \frac{\partial_{[\mu}\sigma\, P_{\nu]}}{\sigma} - \frac{R_{ab\mu\nu}S^{ab}}{4\sigma}, \tag{49}$$

where $X_{[\mu\nu]} := X_{\mu\nu} - X_{\nu\mu}$*; and with* $\tilde{g} = \Omega_{1/2}^2 g$, $\Omega_{1/2}^2 = 1 + \frac{R}{4M^2} - \frac{1}{M^2}\left(\frac{\Box\sqrt{\sigma}}{\sqrt{\sigma}} - \frac{\Lambda}{\sigma}\right)$ *and* $\tilde{u}_\mu = P_\mu/M$,

$$\boxed{M\,\tilde{u}^\nu\tilde{\nabla}_\nu\tilde{u}_\mu = \tilde{u}^\nu F_{\nu\mu}.} \tag{50}$$

The convective flow moves in the conformally rescaled metric like a charge in the spin vorticity. F contains a curvature–spin term with the structure of the Mathisson–Papapetrou coupling, derived here rather than imported. For spin-0 matter $\Lambda = 0$ and $F = 0$, and the law reduces to Proposition 11.4.

Verification. Flat-region identities verified to 10^{-14} on exact superpositions of plane-wave solutions (QGD–112). The curvature coefficients $+\frac{1}{4}$ and $-\frac{1}{4}$, the curved off-shell identity and the curved spin vorticity were verified on an FLRW metric with an arbitrary spinor field, residuals $\leq 6\times10^{-16}$, with the spinor connection constructed from $\nabla\gamma = 0$ (QGD–113). In generic states the spin-vorticity force is 40–70% of the gradient force.

Slow-motion limit (QGD–113). With $\hat{s}$ the spin direction, $\Lambda/\sigma \to \frac{1}{4}\partial\hat{s}\cdot\partial\hat{s}$ (from the exact Fubini–Study identity for unit two-spinors) and $F_{ij} \to -\frac{1}{2}\hat{s}\cdot(\partial_i\hat{s}\times\partial_j\hat{s})$, a Mermin–Ho-type relation, both with $O(v^2)$ convergence verified. The kinetic energy per particle is $p^2/2M - (\hbar^2/2M)\nabla^2\sqrt{n}/\sqrt{n} + (\hbar^2/8M)|\nabla\hat{s}|^2$. For constant spin the law reduces exactly to Chapter 1's irrotational result $\dot{\boldsymbol{p}} = -\nabla(U+Q)$ and to the fixed-spin quantum stress of Section 7. When spin varies, a spin quantum potential and a spin-vorticity force appear that those fixed-spin treatments omit.

Conditions. $\sigma \neq 0$ and P timelike. As for spin-0 matter, the law describes the convective momentum field; reading its integral curves as trajectories is an interpretation.

11.6 Where the natural invariant comes from

Proposition 11.8 (Closure identity (D))**.** *On the static one-null-sector branch* $g = \bar{g} + H\,\mathrm{d}v^2$ *with* $H \to 0$ *at infinity, wherever* $0 < H < 1$*,*

$$\sigma_{\mathrm{amp}} = \sqrt{H} = \beta_{\mathrm{ff}} = \frac{2\zeta_{\mathrm{ff}}}{1+\zeta_{\mathrm{ff}}^2}, \qquad \zeta_{\mathrm{ff}} = \frac{\sqrt{H}}{1+\sqrt{1-H}}, \tag{51}$$

where β_{ff} *is the speed, relative to static observers, of a test particle released from rest at infinity, and* ζ_{ff} *is its Chapter 1 on-shell spinor ratio. Covariantly,* $\sigma_{\mathrm{amp}}^2 = 1 - (u_{\mathrm{ff}}\cdot u_{\mathrm{stat}})^{-2}$*. The sector direction* $k = \mathrm{d}(t+r)$ *is a phase gradient solving the massless limit of Chapter 1's Hamilton–Jacobi equation.*

This extends Chapter 2's Schwarzschild diagnostic to the whole static one-null-sector class. It fails for Kerr: the free-fall speed differs from the Kerr–Schild amplitude, and the oblate k is not closed. It also fails where $FH \neq 1$, for example in Eq. (45). It establishes self-consistency, not origin. The natural invariant's algebraic *form* is derived from the macroscopic spinor equation, while its exterior *values* for a given source are selected by F4.

11.7 Lorentz covariance and gauge dependence

The main equations were tested directly (QGD–110):

Equation or object	Result
Master metric, Eq. (1)	Invariant under frame Lorentz transformations ($T \mapsto \Lambda T$, $\sigma_a \mapsto \sigma_c(\Lambda^{-1})^c{}_a$); tensorial under coordinate changes
Historical $\mathcal{Q}$ with $\bar{\nabla}$	Tensorial: Cartesian and spherical computations agree exactly
Macroscopic spinor equation (Ch. 1 conventions)	Covariant: a spinor representation $S(\Lambda)$ exists and $Su(p)$ solves the equation at Λp
Spinor ratio $\mathsf{R}(p)$	Transforms exactly as $\mathsf{R}(p) \mapsto \mathsf{R}(\Lambda p)$
$\zeta = p/(E + Mc^2)$	Poincaré-ball coordinate of the mass shell; boosts act by Möbius addition (exact). $\beta = 2\zeta/(1+\zeta^2)$ is the Beltrami–Klein coordinate
Spinor component moduli, $\psi^\dagger\psi$	**Not** invariant; $\bar{\psi}\psi$ and $j^\mu j_\mu$ are
Ω_ψ, $T^{\mathcal{Q}}$ (scalar wave)	Covariant
Sector representation σ	**Gauge dependent**: the same Schwarzschild geometry (Kretschmann $12r_s^2/r^6$ in both) has $h = g - \bar{g}$ of rank 1 in the advanced identification and rank 2 in the static one

Three consequences follow:

1. **Spin-$\frac{1}{2}$ version of the quantum terms (QGD–111–113).** The scalar-wave Eqs. (46) and (47) use component amplitudes, which are not invariant for spinors. The covariant spin-$\frac{1}{2}$ amplitude is $(-j \cdot j)^{1/4}$ (Fierz identity, verified). The per-wave level for spin-$\frac{1}{2}$ matter is Theorem 11.7. The universal level holds unchanged: in the energy frame, the stress tensor of a coherent spin-$\frac{1}{2}$ field is $\rho\, u_L u_L + \Pi$ with $\Pi = O((\lambda_C/L)^2)\rho$ (numerical scaling, slope 1.98), and the probability current tilts relative to the energy flow at $O(\lambda_C/L)$ (slope 1.01) through the spin current, which is not an extra stress.

2. **Geometric meaning of the spinor ratio.** The Chapter 1 ratio ζ is not a Lorentz vector. It is a hyperbolic-geometry coordinate (Poincaré ball), and the closure identity (51) is the ball-to-Klein map.

3. **The natural invariant is not a diffeomorphism invariant.** Because the master metric contains a fixed reference $\bar{g}$, the sector representation, including the number of sectors, their rank and their signs, depends on how spacetime is identified with the reference. Only g, and quantities built from it, are gauge invariant. Every QGD prediction under F4 is computed from g, so no prediction is affected, but statements about σ must state the identification used.

12 What the equation makes possible—and what it does not

Equation (43) has an important explanatory role. It expresses effective gravitational equilibrium as a balance of the metric-mediated gravitational action against physical matter stress, projected into the natural-invariant variables. The source is the *complete, conserved* stress of the physical system; a local Dirac-component ratio or a single wave amplitude is not generally an adequate universal source. The Dirac state may contribute through basis-independent currents, energies and stress correlations.

But the action-derived equation also establishes a *limit*: with no direct natural-invariant interactions and no new dynamical operator, the effective field equations contain no distinct propagating σ mode beyond the effective metric degrees of freedom. A new observable could

follow from (i) a well-defined direct coupling between quantum matter and natural-invariant data, (ii) a physically derived independent natural-invariant kinetic/constraint sector, or (iii) a genuinely distinct quantum effective action. None follows inevitably from the signed metric identity; each would require a separately specified covariant action, conservation laws, correct field content and measurable consequences.

Note on route (i) (QGD–100, corrected QGD–105). Research note QGD–100 examined one *hypothetical* realization of route (i): removing the reference term, reading four η-signed sectors as a tetrad, and assuming a microscopic theory with only global frame symmetry. Neither assumption belongs to the formulation of Chapters 1–4. Under them the two-derivative dynamics is New General Relativity [11], whose parameter directions other than the axial one give $\gamma = -2c/(a+b-4c) \neq 1$ at linear order, and whose axial direction is obstructed at nonlinear order according to [12]. This rules out that particular hypothetical route; it says nothing about routes (i)–(iii) in general.

A particularly sharp discriminator is the comparison of different physical quantum source states with matched mean stress and classical boundary data. A measured mean-field difference beyond standard GR plus quantum-field corrections would indicate a new physical coupling. Merely assigning different signed representatives to the same metric cannot change a metric-only observable.

13 Relationship to the archived historical Chapter 4

The historical research program wrote schematically

$$\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu + \cdots . \tag{52}$$

This is *not* equivalent by definition to Eq. (43). The historical Q_μ, G_μ, T_μ expressions have unresolved algebraic/provenance and normalization issues documented in the QGD–052 audit. In particular, no term $\Box_g \sigma_\mu$ or $\Box_g^2 \sigma_\mu$ can be deduced simply from varying the derivative-free composite Einstein–Hilbert action. Adding a standalone σ kinetic term would be a change of the effective theory and requires physical justification, not a cosmetic rewriting of the balance equation. The archived manuscript therefore remains an independent historical source, while the present chapter begins with the precise equation supported by the accepted action and metric map.

14 Conclusion and the next legitimate question

The action-based content of this updated Chapter 4 has one central field-equation result:

$$\boxed{\mathscr{S}^{\rho}_{(s)} = 2 s_s \mathcal{E}_g^{\rho\nu} \sigma_\nu^{(s)} = 0, \qquad \mathscr{S}^{\rho}_{(s)} = \mathscr{G}^{\rho}_{(s)} + \mathscr{T}^{\rho}_{(s)} + \mathscr{Q}^{\rho}_{(s)}.} \tag{53}$$

For the metric-only effective action, $\mathscr{Q} = 0$ and, on the full-rank signed branch, this balance reproduces the established Einsteinian field equations. This is a rigorous mathematical bridge from QGD's natural-invariant variables to effective gravity, not a claim that the underlying physical origin of gravity has been solved. The question that can make QGD more than a geometric reformulation is therefore precise: *Does the quantum matter-wave structure lead to independent, covariantly defined natural-invariant dynamics or source dependence, and does any such effect produce an observable not already predicted by GR and quantum field theory?*

The Newton-calibrated Chapter 3 potential and the controlled $\hbar^2$ quantum-source stress are compatible with this balance, but neither supplies an independent natural-invariant propagation operator. The metric-only quantum-equivalence theorem in Section 10 makes this limitation

exact under its stated rank and action assumptions; the historical attractive 9/2 force would additionally require a nonzero exterior source residual or a changed gravitational response. The historical 9/2 correction remains unverified. Future extensions must derive their covariant operators, coefficient matching and physical observables before assigning a genuinely new wave equation or degree of freedom to σ_μ.

Revision history

QGD–096 established the equivalence/no-independent-correction theorem for the regular metric-only theory and calculated the nonzero vacuum-Poisson residual that the historical attractive 9/2 force would require. **QGD–097** added a separately sourced semiclassical benchmark for a massless four-component Dirac vacuum field: a verified far-field $-2G_N^2 M\hbar/(15\pi c^3 r^3)$ correction to the static Killing-energy potential, taken from published quantum-matter backreaction, not from the historical complex-phase Taylor expansion or a new QGD operator. **QGD–098** independently derived and checked the angular-channel coefficients of the Schwarzschild radial Dirac reduction and recorded why its Liouville-normal effective potential cannot be promoted to a physical force by differentiation. **QGD–101** integrated research notes QGD–099 and QGD–100. **QGD–105** audited those insertions: Remark 3.2 was restricted to Chapter 2's constructed branch; the route-(i) paragraph was restated as a test of one hypothetical realization; and a closing paragraph that reframed QGD's identity was removed, because it did not follow from the chapter's stated assumptions. No earlier equation or label was removed. **QGD–110** added Section 11: the split identity, the quantum term in static variables, the test of the historical derivative trial, the derived quantum conformal factor, the two-level master metric, the closure identity, and Lorentz-covariance and gauge tests. **QGD–111** replaced the open spin-$\frac{1}{2}$ item with covariant results; **QGD–112** added the exact spin-$\frac{1}{2}$ flow law and restricted the name "Dirac" to the microscopic field. **QGD–113** added the curved-metric form of that law (Theorem 11.7), its slow-motion limits, and the cross-reference in Section 7.

Source and research provenance

References

[1] QGD, *Chapter 1: Foundations*, accepted Dirac/matter-wave construction (existing research manuscript).

[2] QGD, *Chapter 2: Field Equations*, complete signed master metric, variational right-inverse and effective action (QGD–075).

[3] QGD, *Chapter 3: Gravitational Normalization and Quantum-Force Verification* (QGD–087 surgical revision; QGD–101 notation update).

[4] QGD–052, *Focused Historical QGT/QGD Field-Equation Provenance Audit*; and QGD–094, *General Natural-Invariant Field Equations and Historical Source Split*, working research checks.

[5] QGD–091–092, *Independent Schwarzschild Dirac radial reduction and referee verification*, conditional test-field coefficient audit (cross-checked in QGD–098).

[6] QGD–090–094, *Quantum Force, Dirac Radial Dynamics, and Quantum-Source Verification*, conditional source and effective-metric studies. Historical $\hbar^2$ operators remain unmatched.

[7] E. D. Carlson, P. R. Anderson, A. Fabbri, S. Fagnocchi, W. H. Hirsch, and S. A. Klyap, "Semiclassical gravity in the far field limit of stars, black holes, and wormholes," *Phys. Rev. D* **82**, 124070 (2010), DOI: 10.1103/PhysRevD.82.124070; see Eqs. (7.2), (8.3), (8.6), and (8.7).

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